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Quotient Groups



Quotient Groups

left cosets of $H < G$

right cosets of $H < G$

$$aH = \{ah : h \in H\}$$

$$Ha = \{ha : h \in H\}$$

Def Let H be a subgroup of G , H is a normal subgroup of G , $H \triangleleft G$, if $aH = Ha$ for all $a \in G$.

Eg) 1) All subgroups of an abelian group are normal.

G abelian

$H < G$

$$aH = \{ah : h \in H\}$$

$$= \{ha : h \in H\}$$

$$= Ha$$

2) Any subgroup of index 2 is normal in G

H, aH two left cosets in H

$$G = H \cup aH \quad \& \quad H \cap aH = \emptyset \Rightarrow H^c = aH$$

H, Ha two right cosets of H in G

$$G = H \cup Ha \quad \& \quad H \cap Ha = \emptyset \Rightarrow H^c = Ha$$

$$\text{So } Ha = aH \quad \& \quad H = H \quad \text{so}$$

left & right cosets are the same.

G/H = set of cosets of H in G

$$G/H = \{H, aH\} \quad \text{where } a \notin H.$$

$$3) \langle r^2 \rangle < D_4 \rightarrow \{e, r^2\}, \{r, r^3\}, \{s, sr^2\}, \{sr, sr^3\}$$

$$\begin{array}{l} s^2 = e \\ r^4 = e \end{array} \left| \begin{array}{l} srs = r^{-1} \\ \text{left cosets } \langle r^2 \rangle \\ \text{right cosets } \{e, r^2\}, \{r, r^3\}, \{s, sr^2\}, \{sr, sr^3\} \end{array} \right.$$

$\langle r^2 \rangle$ $r\langle r^2 \rangle$ $s\langle r^2 \rangle$ $sr\langle r^2 \rangle$

so $r^2 \triangleleft D_4$

$$\begin{array}{l} srs = r^3 \\ sr = r^3s \end{array}$$

$$\langle sr^2 \rangle < D_4$$

left cosets
right cosets

not going to equal

e.g. $rs \langle sr^2 \rangle \neq \langle sr^2 \rangle rs$
so $\langle sr^2 \rangle$ not normal in G .

4) Let $f: G \rightarrow H$ be homomorphism
 then $\ker f \trianglelefteq G$

Previous $f^{-1}(b) = a \ker f$ where $f(a) = b$

Claim Let $b = f(a)$, then $f^{-1}(b) = \ker f a$

$$x \in f^{-1}(b) \Leftrightarrow f(x) = b$$

$$\Leftrightarrow f(x) = f(a)$$

$$\Leftrightarrow f(x) f(a)^{-1} = e_H$$

$$\Leftrightarrow f(x a^{-1}) = e_H$$

$$\Leftrightarrow x a^{-1} \in \ker f$$

$$\Leftrightarrow x \in \ker f a$$

mult on ~~left~~ ^{right}
 by $f(a)^{-1}$

G/H is the set of left cosets of H in G

$H \backslash G$ is the set of right cosets of H in G

Goal: want to give a group structure to G/H
when $H \triangleleft G$.

Def If S & T are subsets of a group G then the product of S & T is $ST = \{s \cdot t : s \in S, t \in T\}$.

$$\underline{\text{Es}}] \quad \langle r^2 \rangle \trianglelefteq D_4 \quad D_4 / \langle r^2 \rangle$$

$$\{e, r^2\} \cdot \underbrace{\{r, r^3\}}_{r \langle r^2 \rangle} = \{ \cancel{e} \cdot \cancel{r}, \cancel{e} \cdot \cancel{r^3}, \cancel{r^2} \cdot \cancel{r}, \cancel{r^2} \cdot \cancel{r^3} \}$$

$$= \{r, r^3\}$$

$$\{r, r^3\} \cdot \{r, r^3\} = \{ \cancel{r} \cdot \cancel{r}, \cancel{r} \cdot \cancel{r^3}, \cancel{r^3} \cdot \cancel{r}, \cancel{r^3} \cdot \cancel{r^3} \}$$

$$= \{e, r^2\}$$

$$\{s, sr^2\} \cdot \{s, sr^2\} = \{e, r^2\}$$

$$\{rs, rs^3\} \cdot \{rs, rs^3\} = \{e, r^2\}$$

Aside

$$D_4 / \langle r^2 \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$$

Klein group

Prop 13.7

Let $H \leq G$ then the following conditions are equivalent

1) $H \triangleleft G$

2) $(aH)(bH) = (ab)H \quad \forall a, b \in G$

3) $(Hs)(Hb) = H(sb) \quad \forall s, b \in G$

4) the product of two left cosets is again a left coset

5) the product of two right cosets is again a right coset.

1 $\Rightarrow$ 2

Suppose $H \triangleleft G$ want to show: $(aH)(bH) = (ab)H$

$$(aH)(bH) = \{a \underline{h} b h' : \text{where } h', h \in H\}$$

$hb \in Hb$ but $Hb = bH$
so $hb \in bH \Rightarrow hb = bh''$ for some $h'' \in H$

$$= \{ab h'' h' : \text{where } h'', h' \in H\}$$

$$= (ab)H$$

$2 \Rightarrow 1$ Suppose $(eH)(bH) = abH \quad \forall a, b \in G$

Let $a = e$ then $eHbH = HbH$
 $\parallel$
 bH

So $HbH = bH$ but $Hbe \subseteq HbH$

$\Rightarrow Hb \subseteq bH$

Consider $(eH)(b^{-1}H) = Hb^{-1}H$

$\parallel$
 $b^{-1}H$

$Hb^{-1}e \subseteq b^{-1}H$

$\Rightarrow Hb^{-1} \subseteq b^{-1}H$

$\Rightarrow bH \subseteq Hb$

mult by
 b on both
sides

so $Hb = bH$.
 $\Rightarrow H \trianglelefteq G$

Def Let $N \triangleleft G$ then the set of cosets of N in G

G/N is a group under the multiplication

$$aN \cdot bN = (ab)N.$$

This is called the quotient group of G modulo N .

Eg

$$D_4 / \langle r^2 \rangle = \left\{ \overset{\langle r^2 \rangle}{\underset{||}{\{e, r^2\}}}, \{s, sr^2\}, \{r, r^3\}, \{r, sr^3\} \right\}$$

so $D_4 / \langle r^2 \rangle \overset{C_{D_4/\langle r^2 \rangle}}{=} \mathbb{Z}_2 \times \mathbb{Z}_2$

First Isomorphism Thm Let $f: G \rightarrow H$ be a group hom.

then $\bar{f}: G/\ker f \rightarrow \text{im } f$ by $\bar{f}(a \ker f) = f(a) \forall a \in G$

is a group isomorphism.

PS well defined suppose $a \ker f = a' \ker f$ NTS $f(a) = f(a')$

$$f(a) = f(a') \Leftrightarrow f(a)^{-1} f(a') = e_H$$

$$\Leftrightarrow f(a^{-1} a') = e_H$$

$$\Leftrightarrow a^{-1} a' \in \ker f$$

$$\Leftrightarrow a' \in a \ker f$$

$$\Leftrightarrow a' \ker f = a \ker f$$

homomorphism $\bar{f}(a \ker f \cdot b \ker f) = \bar{f}(ab \ker f)$

$$= f(ab) = f(a)f(b)$$

$$\bar{f}(a \ker f) \cdot \bar{f}(b \ker f) = f(a) \cdot f(b)$$

isomorphism
(bijections)

$$\begin{aligned} \ker \bar{f} &= \{ a \ker f : \bar{f}(a \ker f) = e_H \} \\ &= \{ a \ker f : f(a) = e_H \} \\ &= \{ a \ker f : a \in \ker f \} \\ &= \{ \ker f \} = e_{G/\ker f} \quad \square \end{aligned}$$

$\mathbb{M}$

master group

$$|\mathbb{M}| = 196883$$

$$\langle s_1, s_2 \rangle$$

